Bellman Meets Kolmogorov: Unsupervised Reinforcement Learning Through Mastering Chaos

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Abstract

Reinforcement learning (RL) is a powerful paradigm for training agents, yet its success rests on domain expertise of human engineers who design informative reward signals for every new task. Unsupervised RL aims to replace this engineering with intrinsic motivation (IM): reward signals that emerge from the agent-environment interaction itself. Existing IM objectives, however, reward state space coverage or model surprise without providing a strong inductive bias towards useful behaviors. We introduce **Forward CIP** (Controllable Information Production), an objective derived from Kolmogorov-Sinai entropy that is directly compatible with existing RL algorithms. Maximizing this quantity alone results in unsupervised discovery of skills such as balancing and maintaining controllability, which are essential for more complex robot behaviors. Paired with a minimal forward-velocity reward, our method produces coordinated gaits such as hopping and running which otherwise require reward engineering to learn.

1 Introduction

The field of intrinsic motivation (IM) seeks to identify the goal-agnostic principles of self-motivation in biological systems and to replicate them in artificial ones. One such principle is the *edge of chaos* (EOC) (Zhang et al. 2025b; Tower 2024; Wissner-Gross and Freer 2013): the hypothesis that intelligence arises from an agent's ability to operate at the boundary between stable and chaotic dynamics. In physically embodied agents, this boundary corresponds to states which are maximally unstable yet still controllable through active regulation. The upright stance of a humanoid is the canonical example: it affords walking, running, and hopping, yet it is also the most precarious configuration, since small mistakes readily lead to collapse. The EOC is thus distinct from pure chaos, where perturbations grow faster than any policy can suppress. In this work, we make the EOC directly optimizable as a reinforcement learning (RL) objective.

A plethora of IM objectives have been proposed, yet they are employed almost exclusively in auxiliary roles: as exploration bonuses added to an extrinsic reward (Raileanu and Rocktäschel 2020; Pathak et al. 2017; Burda et al. 2018), or as pre-training signals whose value is only realized once a downstream task is imposed (Liu and Abbeel 2021; Eysenbach et al. 2019; Laskin et al. 2021). In neither role does the objective require that the behaviors it produces be useful in their own right: curiosity rewards vanish as the learned model improves, and diversity objectives are indifferent to whether the states being explored are dynamically trivial or rich. In both cases the objective is a heuristic designed by human intuition rather than derived from dynamical systems theory, lacking a principle (such as the EOC) that distinguishes significant states from insignificant ones.

Recently Shah and Tiomkin (2026) introduced Controllable Information Production (CIP), an IM objective that closes this gap. CIP is grounded in Kolmogorov-Sinai Entropy (KSE) (Kolmogorov 1958; Sinai 1959), a global property of dynamical systems that governs the rate (in $\mathrm{bit}/\mathrm{sec}$) at which the dynamics generate new distinguishable trajectories. Whereas visitation entropy weights all states alike, KSE weights states by the complexity of the dynamics passing through them, and is therefore maximized not by uniform state coverage but by occupying the most dynamically sensitive regions. A controller maximizing CIP was shown to drive agents to EOC regimes across a suite of benchmark tasks, including raising a humanoid-like system from a passive configuration to an upright stand. This demonstrated that IM grounded in a principled dynamical quantity can serve not merely as an exploration bonus, but rather as the primary objective of the agent.

Existing optimizers of CIP, however, are sampling-based MPC planners (Howell et al. 2022; Pinneri et al. 2021; Williams, Aldrich, and Theodorou 2015) which are ill-suited to the CIP objective. Such planners must forward-simulate a large number of candidate trajectories at every timestep and select the best action sequences. Because evaluating CIP requires running a Riccati recursion over the local Jacobians of each rollout, the cost of every discarded candidate trajectory is far higher than under a standard per-state cost. Gradient-based planners (Zhang et al. 2025a; Tassa, Erez, and Todorov 2012) avoid the need for many rollouts and instead require a differentiable cost evaluated around a local trajectory. Since the CIP objective already depends on the state Jacobian, differentiating it for planning would demand second-order derivatives of the dynamics, which are impractical to obtain in high-dimensional systems.

RL (Sutton, Barto et al. 1998) is a natural alternative to

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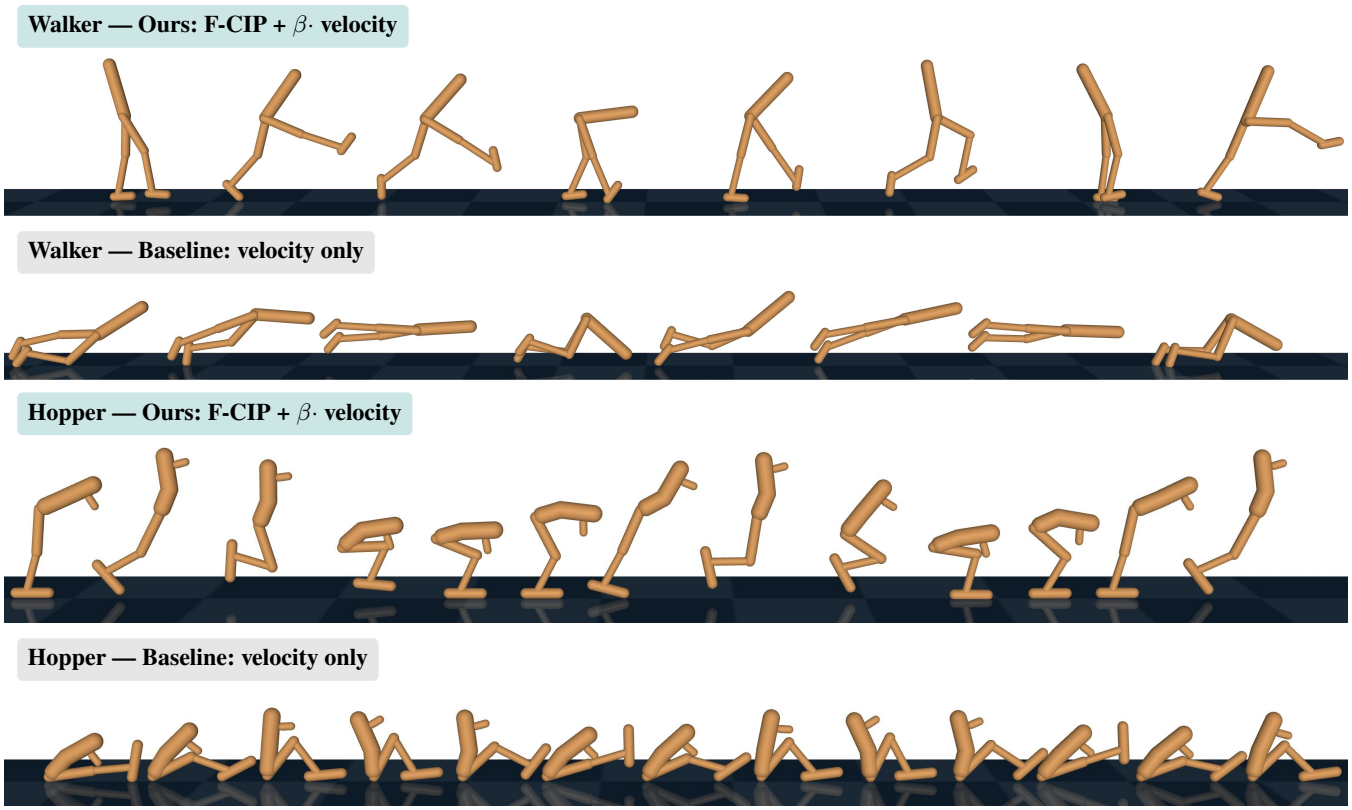


Figure 1: **Emergent locomotion strategies on walker and hopper.** F-CIP pretraining followed by fine-tuning on F-CIP plus a β-weighted forward-velocity reward yields upright running and hopping (teal), whereas the velocity reward alone collapses to bouncing and scooting along the ground (gray).

MPC: it amortizes the sampling cost into a learning phase and delivers a reactive feedback policy. The resultant policy is faster to use at test time because it can be run without expensive re-planning at each step. Yet RL imposes two structural requirements on its objective: (i) the reward must be emitted incrementally during the forward agent-environment interaction, and (ii) the reward must be Markovian in the agent's state to support bootstrapped value learning. CIP in its original form violates both. It is computed by a backward-in-time Riccati recursion: the increment at the current time depends on a matrix propagated from the terminal time, i.e., on the entire future trajectory.

We introduce **Forward CIP** (F-CIP)[1]: a forward-in-time incremental decomposition of CIP that makes it, for the first time, directly optimizable by RL. This resolves requirement (i) by exposing an instantaneous per-step reward, and requirement (ii) by identifying the information matrix S_t as a sufficient statistic of the past: augmenting the state with S_t exactly restores the Markov property (Proposition 3). We prove (Lemma 1, Theorem 4) that the forward iteration converges to the same quantity as the backward recursion, so no approximation is introduced. Empirically, PPO (Schulman et al. 2017) on the F-CIP reward alone yields unsupervised self-righting and standing. With a single additional velocity term, agents learn upright hopping and running; the same velocity reward without F-CIP collapses to degenerate gaits.

Our contributions are summarized as follows:

1. **Theory.** A forward-in-time characterization of CIP as a sum of per-step increments (Lemma 1).
2. **Algorithm.** A numerically stable log-domain iteration exposing a per-step reward and an information-matrix memory S_t, casting CIP maximization as a standard RL problem.
3. **Experiments.** The first RL agent trained directly on the CIP objective. Maximizing it drives the unsupervised discovery of standing behaviors, while combining it with a downstream objective yields coherent locomotion strategies such as hopping and running which otherwise require reward engineering to learn.

2 Related Work

Existing IM objectives fall broadly into four families: curiosity, skill discovery, state-entropy maximization, and EOC. We review each and position CIP against them.

Curiosity. Curiosity-driven methods reward the surprise of a learned dynamics model, measured as prediction er-

[1] *Throughout, CIP refers to the underlying quantity (the open-loop KSE) while F-CIP refers to our per-step decomposition and the reward it exposes; the two coincide in value by Theorem 4.*



Figure 2: **Unsupervised standing up behavior via F-CIP maximization**. This behavior is achieved through maximization of F-CIP alone without any extrinsic reward.

ror (Pathak et al. 2017; Burda et al. 2019), error of a random target network (Burda et al. 2018), or ensemble disagreement (Pathak, Gandhi, and Gupta 2019). These objectives are defined relative to the deficiencies of the learned model rather than a property of the environment itself. Consequently, the reward signal can vanish as the model improves (Raileanu and Rocktäschel 2020), and it is maximized by irreducibly unpredictable stimuli regardless of their behavioral relevance, e.g., the noisy-TV problem (Burda et al. 2018). CIP instead rewards a dynamical invariant of the system: its signal neither vanishes with learning progress nor depends on model error.

Skill-Discovery. Skill-discovery methods (Zheng et al. 2025) maximize the mutual information between a latent skill variable and the states that the agent visits under that skill (Eysenbach et al. 2019; Sharma et al. 2019; Gregor, Rezende, and Wierstra 2016). This principle, however, is agnostic to where the state space is behaviorally significant. For example a diverse set of trivially stable skills that partition the state space satisfies the objective as well as a set of dynamically rich ones. CIP takes an alternative approach: it concentrates the agent at the states of maximal instability rather than spreading it over the entire state space.

State-entropy maximization. A related family directly maximizes the entropy of the state-visitation distribution (Liu and Abbeel 2021; Lee, Sun, and Abbeel 2025). Although both entropy over the state-visitation distribution and KSE are types of entropies, they measure different things: visitation entropy is a static property of the state-visitation distribution, whereas KSE is the entropy production rate of the dynamical system itself measured on the optimal partition of the state space (as described in Section 3.1). The two are independent: an agent can achieve maximal visitation entropy while residing mostly in regions of low KSE, and conversely can hold a single EOC state of high KSE with near-zero visitation entropy. Our objective is the latter.

Edge of Chaos. The closest objectives to our work are also based on the EOC principle. Empowerment (Klyubin, Polani, and Nehaniv 2005) is the channel capacity from the agent's actions to its future states; it generalizes the controllability Gramian of linear systems and has been related to the finite-time Lyapunov exponent spectrum (Tiomkin et al. 2024). Empowerment, however, has resisted exact factorization into a sum of rewards: channel capacity over a multi-step horizon is a supremum over action distributions that does not separate into per-step increments. Consequently, RL with empowerment has relied on variational lower bounds and fixed-horizon estimators (Karl et al. 2022). CIP shares the EOC principle but is grounded in KSE, and was previously optimized only by sampling-based MPC for a similar reason: its backward recursion admits no online reward. The present work removes this obstruction. To our knowledge, F-CIP is the first dynamics-grounded intrinsic RL objective with an exact per-step decomposition, requiring neither a variational bound nor a fixed optimization horizon.

3 Preliminaries

3.1 Kolmogorov-Sinai Entropy

Kolmogorov-Sinai entropy (KSE) quantifies the rate at which a dynamical system $f : \mathcal{X} \to \mathcal{X}$ produces new, distinguishable trajectories (Kolmogorov 1958; Sinai 1959; Walters 1982; Petersen 1983), and is strictly positive for chaotic dynamics. A finite measurable partition $\xi = \{P_1, \ldots, P_k\}$ of state space $\mathcal{X}$ acts as a coarse measurement: each step of the dynamics splits cells of ξ into finer pieces, so longer horizons distinguish more trajectories.

$$h_{\mathrm{ks}}(f) := \sup_{\xi} \lim_{T\to\infty} \frac{1}{T} H\left(\bigvee_{t=0}^{T-1} f^{-t}\xi\right).$$

Here $H(\cdot)$ denotes the Shannon entropy of the partition ξ, $f^{-t}\xi$ its preimage under t steps of the dynamics, and $\bigvee$ denotes the join of partitions. As T increases, the join $\bigvee_{t=0}^{T-1} f^{-t}\xi$ refines ξ: two points share a cell only if their trajectories visit the same sequence of cells of ξ for all T steps. Where the dynamics are sensitive to initial conditions, nearby trajectories separate, so the cells of the join become high resolution exactly in these regions. KSE is the supremum, over all finite partitions, of the per-step entropy growth rate measured on the joined partition.

The partition-based definition is rarely tractable due to the supremum over ξ (Murphy and Bassett 2024). It can instead be calculated by the sum of positive Lyapunov exponents (LEs) of the system, denoted as λ_i, which are based on sensitivity analysis (Wolf et al. 1985). Pesin's theorem (Pesin

1977) formalizes this connection:

$$h_{\text{ks}}(f) = \sum_{\lambda_i > 0} \lambda_i. \tag{1}$$

Despite this identification of KSE with the sum of positive LE it has still not been shown to decompose into a sum of Markovian rewards. Our work bridges this gap by providing a forward incremental accumulation of instantaneous quantities which converges to the system's KSE.

3.2 Controllable Information Production

CIP is defined as the KSE of the open-loop dynamical control system $f : \mathcal{X} \times \mathcal{U} \to \mathcal{X}$ where $\mathcal{U}$ is the action space. Open-loop KSE is the entropy production rate which the agent is exposed to along its trajectory, considering that infinitesimal perturbations grow without regulation due to feedback by a policy $\pi : \mathcal{X} \to \mathcal{U}$.

$$\text{CIP} := h_{\text{ks}}(f^{\mathbf{ol}}), \qquad f^{\mathbf{ol}}(x) = f(x, \mathbf{sg}(\pi(x)))$$

The symbol $\mathbf{sg}$ denotes the stop-gradient operation. It was shown that CIP can be calculated via a backward recursion:

$$\begin{aligned} h_t &= h_{t+1} + \log\det(Y_{t+1} + f_{x_t} f_{x_t}^\top), & h_T &= 0, \\ Y_t &= \mathbf{I} - f_{x_t}^\top (Y_{t+1} + f_{x_t} f_{x_t}^\top)^{-1} f_{x_t}, & Y_T &= \mathbf{I}, \end{aligned} \tag{2}$$

where $f_{x_t} := \frac{\partial f}{\partial x}|_{(x_t, u_t)}$ is the local Jacobian with respect to state. If h_t is propagated backward in time along an infinitely long trajectory then its one-half time average converges to the open-loop KSE:

$$\lim_{T\to\infty} \frac{1}{2T\Delta t} h_0 = h_{\text{ks}}(f^{\mathbf{ol}}). \tag{3}$$

The increment between h_t and h_{t+1} in Equation (2) is not available to the agent during the forward rollout as a reward quantity: the source of CIP's incompatibility with RL. Furthermore, it depends on Y_{t+1}, which is a function of the entire future trajectory: breaking the Markov Property required for bootstrapped value learning. In the next section we present a dual forward-in-time calculation of CIP which enables training with RL.

4 Forward Estimation of CIP

We factor the calculation of CIP into a sum of increments emitted during the forward agent-environment interaction, so that each increment can serve as an instantaneous reward. Our key insight comes from the control-estimation duality (Todorov 2008): the backward Riccati recursion in Equation (2) is a control-side object, and its estimation-side dual is a forward-in-time covariance iteration whose increments depend only on the current Jacobian f_{x_t} and a propagated information matrix S_t. We develop this in four steps: the forward iteration converges to the same open-loop KSE as the backward recursion (Lemma 1); a log-domain form of the iteration is numerically stable and exposes the per-step reward (Lemma 2); the augmented state $z_t := (x_t, S_t)$ defines a legitimate MDP (Proposition 3); and the return of that MDP is exactly CIP (Theorem 4).

The uncertainty in the agent's state is given by a covariance matrix Σ_t. The time evolution of this uncertainty is determined by the local Jacobian as well as the covariance of the process noise W_{t+1} injected at each step. The following lemma shows that its log-volume growth rate recovers CIP exactly.

Lemma 1 (Forward covariance characterization of CIP)**.** *Let*

$$\Sigma_{t+1} = W_{t+1} + f_{x_t} \Sigma_t f_{x_t}^\top, \qquad \Sigma_0 = W_0, \tag{4}$$

be the open-loop covariance propagation equation for state uncertainty, where W_t is the process-noise covariance injected at step t, uniformly bounded and positive definite. Then:

$$\lim_{T\to\infty} \frac{1}{2T\Delta t} \ln\det \Sigma_T = h_{ks}(f^{\mathbf{ol}}) = \text{CIP}.$$

Proof. Unrolling Equation (4) gives a grammian matrix:

$$\Sigma_T = \sum_{t=0}^{T} \Phi_{t:T}\, W_t\, \Phi_{t:T}^\top,$$

with $\Phi_{t:T} = \prod_{k=t}^{T-1} f_{x_k}$ as a product of Jacobians. Reindexing $\tau = T - t$ and setting $\Psi_\tau := \Phi_{T-\tau:T}^\top$ casts this as the cumulative-sensitivity sum $\Sigma_T = \sum_{\tau=0}^{T} \Psi_\tau^\top W_{T-\tau} \Psi_\tau$, where each Ψ_τ is a length-τ product of trajectory Jacobians. Because Ψ_τ and $\Phi_{T-\tau:T}$ share singular values, the products exhibit the exponential growth $\|\Psi_\tau v_i\| \sim e^{\tau \Delta t \lambda_i^{\mathbf{ol}}}$ governed by Lyapunov Exponents $\{\lambda_i^{\mathbf{ol}}\}$. This is exactly the object of (Shah and Tiomkin 2026, Lemma A.2), whose weighting invariance (Shah and Tiomkin 2026, Lemma A.3) admits any uniformly bounded positive-definite $\{W_t\}$; together they give $\frac{1}{2T\Delta t} \ln\det \Sigma_T \to \sum_{\lambda_i^{\mathbf{ol}} > 0} \lambda_i^{\mathbf{ol}}$, which Pesin's theorem (Pesin 1977) identifies with KSE and hence CIP. □

Direct propagation of Equation (4) is numerically unstable, since the condition number of Σ_t grows exponentially with time. Furthermore, there is no additive increment available to interpret as a reward. To overcome this instability we instead propagate Σ_t in the $\ln\det$ domain via an iteration derived in Lemma 2 which exposes an instantaneous reward signal.

Lemma 2 (Log-domain iteration)**.** *Let $h_t := \ln\det \Sigma_t$ and let $S_t := \Sigma_t^{-1}$ be the information matrix, and choose $W_t = \mathbf{I}$ (admissible by the weighting invariance of Lemma 1). Then:*

$$h_{t+1} = h_t + \ln\det\left(S_t + f_{x_t}^\top f_{x_t}\right), \qquad h_0 = 0, \tag{5}$$

$$S_{t+1} = \mathbf{I} - f_{x_t}\left(S_t + f_{x_t}^\top f_{x_t}\right)^{-1} f_{x_t}^\top, \qquad S_0 = \mathbf{I}. \tag{6}$$

Proof. With $W_t = \mathbf{I}$, Equation (4) reduces to $\Sigma_{t+1} = \mathbf{I} + f_{x_t} \Sigma_t f_{x_t}^\top$ with $\Sigma_0 = \mathbf{I}$. Taking $\ln\det$ and applying Sylvester's identity yields the first equation; inverting and applying the Woodbury identity yields the second. □

The iteration in Lemma 2 is numerically stable: because $\Sigma_t \succeq \mathbf{I}$ for all t, the information matrix is bounded, $0 \preceq S_t \preceq \mathbf{I}$, so the exponential growth of Σ_t underflows towards zero in S_t, remaining well-conditioned as a neural network input.

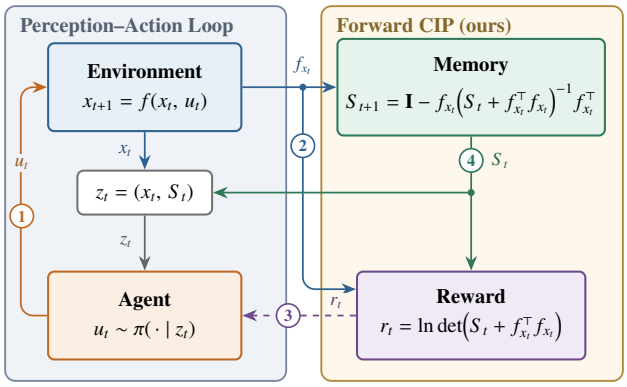


Figure 3: **Schematic depiction of our method**. The agent interacts with the environment, which returns a local Jacobian $f_{x_t} := \frac{\partial f}{\partial x}|_{(x_t,u_t)}$ used to emit a reward r_t and update the agent's memory S_t. Note that the memory is updated after the reward emission so the reward r_t is calculated with the current memory S_t.

The increment in Equation (5) defines a per-step reward:

$$r(x_t, u_t, S_t) := \ln\det\left(S_t + f_{x_t}^\top f_{x_t}\right) = h_{t+1} - h_t. \quad (7)$$

The reward carries the memory term S_t in its signature, so it is not Markovian in x_t alone. The following proposition shows that S_t is a sufficient statistic of the past: augmenting the state with it recovers a legitimate Markov decision process as stated in the following proposition.

Proposition 3 (Markov restoration)**.** *Let $z_t := (x_t, S_t) \in \mathcal{Z} := \mathcal{X} \times \mathcal{S}$, where $\mathcal{S}$ denotes the symmetric matrices satisfying $0 \preceq S \preceq \mathbf{I}$. Then the tuple $(\mathcal{Z}, \mathcal{U}, p_z, r)$, with r as in Equation (7), is a Markov decision process.*

Proof. The transition of z_t factorizes: $x_{t+1} \sim p(\cdot \mid x_t, u_t)$ is the environment step, and S_{t+1} in Equation (6) is a deterministic function of (x_t, u_t, S_t) through the state Jacobian f_{x_t}. The reward $r(z_t, u_t)$ depends only on the current augmented state and action. Hence z_{t+1} and r_t are determined by (z_t, u_t) alone. □

In the following theorem, we state the main result: the return of this MDP is exactly the CIP estimator of Lemma 1.

Theorem 4 (Exact decomposition of CIP)**.** *In the MDP of Proposition 3, the undiscounted return telescopes, $\sum_{t=0}^{T-1} r(z_t, u_t) = \ln\det\Sigma_T$, and consequently*

$$\lim_{T\to\infty} \frac{1}{2T\Delta t} \sum_{t=0}^{T-1} r(z_t, u_t) = h_{ks}(f^{\mathbf{ol}}). \quad (8)$$

Maximizing the average reward of the MDP is therefore equivalent to maximizing CIP.

Proof. By Lemma 2, $\sum_{t=0}^{T-1} r_t = h_T - h_0 = \ln\det\Sigma_T$. The limit follows from Lemma 1. □

Algorithm 1: Training Forward CIP and Extrinsic Reward

Require: policy π_θ, value function V_ϕ, iterations $T_{\text{pre}}, T_{\text{adapt}}, T_{\text{warm}}$, extrinsic weight $\beta_{\max}$, horizon H, N environments

for $i := 1, \ldots, T_{\text{pre}} + T_{\text{adapt}}$ **do**
  $\beta \leftarrow \beta_{\max} \cdot \min\left(1, \frac{\max(0, i - T_{\text{pre}})}{T_{\text{warm}}}\right)$
  for each of N parallel environments **do**
    $x_0 \sim \rho_0$, $S_0 \leftarrow \mathbf{I}$
    for $t := 0, \ldots, H-1$ **do**
      $z_t \leftarrow (x_t, S_t)$ — Prop. 3
      Take action $u_t \sim \pi_\theta(\cdot \mid z_t)$
      Receive x_{t+1} and f_{x_t}
      $r_t \leftarrow \ln\det\left(S_t + f_{x_t}^\top f_{x_t}\right)$ — Eq. (7)
      $S_{t+1} \leftarrow \mathbf{I} - f_{x_t}(S_t + f_{x_t}^\top f_{x_t})^{-1} f_{x_t}^\top$ — Eq. (6)
      $\hat{r}_t \leftarrow r_t + \beta\, v_t$
    end for
  end for
  Update θ, ϕ on $\{(z_t, u_t, \hat{r}_t)\}$ — Standard PPO
end for

Implementation. Theorem 4 is an average-reward statement, whereas standard policy-gradient algorithms optimize a discounted objective. In practice we optimize the discounted proxy with γ close to 1; the discounted-optimal policy converges to the average-reward-optimal policy as $\gamma \to 1$.

The augmented state $z_t = (x_t, S_t)$ is passed to both the value function and the policy. As S_t is symmetric, only its $n(n+1)/2$ unique entries need be carried, and its boundedness in $(0, \mathbf{I}]$ makes it a well-conditioned network input despite the augmentation. The Jacobian f_{x_t} is obtained from the Mujoco-MJX (Todorov, Erez, and Tassa 2012) simulator at each step. The pseudocode in Algorithm 1 summarizes the training procedure.

5 Experiments

Our experiments address three questions: **Q1.** Does a learned value function converge to the true KSE on a known attractor? **Q2.** What behaviors are induced by maximizing F-CIP, with no extrinsic reward? **Q3.** Can F-CIP simplify reward engineering on locomotion tasks?

Q1. KSE Estimation We validate the convergence of our learned incremental estimator on the Lorenz attractor (Lorenz 2017) with a known KSE value of 0.906 nats/s (Sprott 1997). The KSE is extracted from the learned estimator as $\frac{1-\gamma}{2\Delta t}\mathbb{E}[V_\phi(z_t)]$, where the prefactor converts the accumulated discounted return into a rate. Here V_ϕ is estimated by TD learning (Sutton, Barto et al. 1998) on trajectories sampled from the attractor. Figure 4a demonstrates that the estimator converges to the known value for a range of discount factors γ. Setting γ to zero still enables estimation of KSE since that special case corresponds to the value function learning only the instantaneous reward. However, we observe that increasing values of γ result in faster convergence to the true KSE. We visualize the output of the value function over

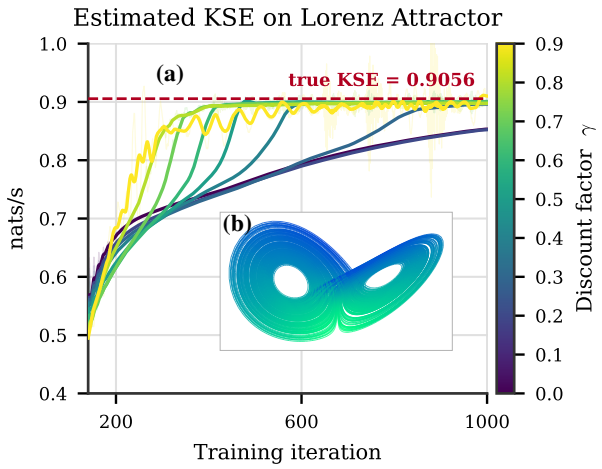


Figure 4: **Learning the KSE on the Lorenz attractor**. **(a)** convergence of the learned value function to the known KSE for a range of $\gamma \in [0.0, 0.9]$; **(b)** the attractor colored by the learned $V_\phi(z)$ for $\gamma = 0.9$ where bright green indicates higher value.

a trajectory of the attractor for $\gamma = 0.9$ in Figure 4b where bright green coloration corresponds to larger values. Since the value function learns a discounted summation its output can be interpreted as a local indicator of the "chaos-to-go". The value function attains its largest values near the splitting point between the two attractor wings, where the local dynamics are most unstable and trajectories can diverge towards either wing. This property motivates using the learned value as a local signal of proximity to the EOC.

Q2. Emergent Behaviors Maximizing F-CIP alone, with no extrinsic reward, produces swing-up and upright stabilization: Figure 2 shows a full swing-up and stabilization trajectory in the gibbon environment. Table 1 quantifies this across all four systems, reporting the normalized height of each system's extremity (i.e. pendulum tip; gibbon head) over the final 10% of the episode, averaged over 10 seeds. All agents start from the hanging pose, so the intrinsic objective alone must drive both swing-up and stabilization. F-CIP is the only objective that reliably reaches the upright configuration on all four systems. The baselines use the default parameters of the Unsupervised Reinforcement Learning Benchmark (Laskin et al. 2021) which implements all objectives with DDPG (Lillicrap et al. 2019); the skill-based methods (DIAYN and SMM) are scored on their best skill, i.e., the one attaining the highest average height. Finally, since the F-CIP objective in Equation (8) is an average-reward quantity, we also train an average-reward PPO variant (APPO) (Ma et al. 2021), which matches the reliability of standard PPO.

Q3. Reward Simplification Inducing coherent locomotion strategies such as hopping and walking using RL usually requires careful reward engineering. The simplest reward of maximizing forward velocity describes the desired outcome to move forward but imposes no bias on how that should

System	Intrinsic Motivation Objective					
	F-CIP (PPO)	F-CIP (APPO)	DIAYN	SMM	ICM	APT
CP	**0.99**	**0.99**	0.26	0.68	0.07	0.64
DP	**0.99**	**0.99**	0.44	0.06	0.05	0.25
TP	**0.99**	**0.99**	0.38	0.14	0.09	0.48
G	**0.98**	**0.99**	0.51	0.52	0.05	0.41

Table 1: **Comparison of IM on pendulum environments**. We report mean extremity height over the final 10% of the episode averaged over 10 random seeds. Values are min-max normalized so that 0.0 corresponds to the initial hanging configuration and 1.0 corresponds to the fully upright pose. The abbreviations CP, DP, TP, and G correspond to cart pole, double pendulum, triple pendulum, and gibbon, respectively. The comparison is not algorithm-matched: F-CIP is optimized with PPO/APPO, whereas the baselines use the reference URLB (Laskin et al. 2021) implementations, which are built on DDPG.

be achieved. Maximizing a forward velocity reward with an RL agent results in degenerate solutions such as "scooting" along the ground (gray rows of Figure 1) rather than adopting an upright gait. The standard remedy is to design a reward to incentivize upright posture, requiring domain knowledge about the robot itself. We show that CIP can supply the inductive bias to maintain an upright posture, eliminating the need for this domain knowledge.

Our procedure has two phases (Algorithm 1). We first pre-train the agent for T_{pre} steps to maintain a stable and upright posture using the F-CIP reward (Equation (7)). Next, we adapt the pre-trained policy for T_{adapt} steps by exposing it to a forward velocity reward weighted by a β parameter, which warms up from zero towards $\beta_{\max}$ over T_{warm} steps. The agent optimizes the combined objective: $\hat{r}_t = r_t + \beta v_t$, where v_t is the forward velocity component. Crucially, our method never optimizes the oracle hand-designed reward; that reward is only used at test time to evaluate task return.

Table 2 reports rollout statistics for both tasks. All baselines collapse to scooting: they hold a low torso height and accrue near-zero task return. Our method instead learns to hop and to run (teal rows of Figure 1), maintaining an upright posture and recovering the majority of the oracle's task return despite never observing that reward.

On hopper, the large variance of our task return is an artifact of the oracle metric rather than of the gait: the oracle reward is gated to zero whenever the torso leaves a narrow height band, and while all 10 evaluation seeds hop upright, some hop at heights outside this band and score zero despite a correct gait. Notably, our hopping policy travels $\sim 40\%$ faster than the oracle policy (1.47 vs. 1.04 m/s), suggesting that the hand-designed height band could be overly restrictive: it penalizes the torso height excursions that fast hopping requires. On walker, our gait falls within the oracle's height band, yielding correspondingly low variance. Its forward velocity is comparable to the fastest baselines, yet ours is the only method besides the oracle to achieve an upright running

Method	Hopper			Walker		
	Task Return ↑	Fwd. Vel. (m/s)	Height (m)	Task Return ↑	Fwd. Vel. (m/s)	Height (m)
Ours	$\mathbf{0.39 \pm 0.38}$	1.48 ± 0.33	0.73 ± 0.15	$\mathbf{0.61 \pm 0.03}$	4.52 ± 0.22	1.19 ± 0.04
Fwd. Vel. only	0.00 ± 0.00	0.77 ± 0.17	0.33 ± 0.03	0.10 ± 0.04	4.41 ± 0.22	0.47 ± 0.10
ICM†	0.00 ± 0.00	1.19 ± 0.03	0.28 ± 0.02	0.06 ± 0.01	4.10 ± 0.17	0.35 ± 0.03
APT†	0.00 ± 0.00	1.36 ± 0.06	0.29 ± 0.04	0.10 ± 0.01	4.63 ± 0.18	0.47 ± 0.04
DIAYN†	0.00 ± 0.00	1.36 ± 0.05	0.28 ± 0.02	0.06 ± 0.01	2.00 ± 0.21	0.43 ± 0.04
SMM†	0.00 ± 0.00	1.66 ± 0.06	0.22 ± 0.02	0.11 ± 0.02	4.26 ± 0.16	0.51 ± 0.04
Full reward *(oracle)*	0.52 ± 0.12	1.04 ± 0.23	0.79 ± 0.09	0.78 ± 0.04	6.07 ± 0.33	1.13 ± 0.02

Table 2: **Reward simplification on Hopper and Walker.** Mean and standard deviation calculated over 10 evaluations of the trained policies. Ours pre-trains with F-CIP alone, then fine-tunes on the forward-velocity reward while continuing to maximize F-CIP (Algorithm 1). Task return is the comparison metric and is measured with the full hand-designed reward; **bold** marks the best performance among the methods. Forward velocity and torso height are reported as diagnostics of the emergent gait. Each pretraining baseline (†) is run in its intended URLB (DDPG) configuration: the intrinsic objective is maximized during pretraining and then replaced by the velocity reward during fine-tuning. In contrast, Ours continues to maximize F-CIP throughout fine-tuning. Ours, Fwd. Vel. only, and Full reward *(oracle)* are trained with the same PPO implementation, so the Fwd. Vel. only row serves as an algorithm-matched control.

gait whereas the baselines attain their speed by scooting.

6 Discussion

We introduced F-CIP, a practical scheme for maximizing CIP with modern RL. Whereas CIP was previously optimizable only by expensive MPC planners, our exact per-step decomposition makes it compatible with bootstrapped value learning: the learned value function recovers the true KSE of a known chaotic attractor (Q1), maximizing the F-CIP reward alone produces swing-up and upright stabilization where standard IM objectives, in their reference URLB configurations, do not (Q2), and F-CIP pre-training replaces the postural reward engineering that locomotion tasks otherwise require (Q3).

These results support a broader position: intrinsic motivation need not be confined to an exploration bonus. Traditionally in continuous control, IM objectives serve almost exclusively as auxiliary signals. This aligns with their underlying design, as principles like coverage and surprise are inherently geared toward exploration. CIP is a different IM signal: because it targets EOC states directly, it produces useful behavior in both roles. As a primary objective it discovers self-righting with no external signal; as an auxiliary term it supplies an inductive bias toward controllable instability that reward engineers otherwise encode by hand.

Our method has three main limitations. First, the reward requires analytical Jacobians. Differentiable simulators such as MuJoCo-MJX (Todorov, Erez, and Tassa 2012) provide these efficiently, and F-CIP is well suited to this setting; physical reality, however, offers no Jacobians, so sim-to-real transfer will require either training entirely in simulation or learning the Jacobians from data. Second, the memory matrix S_t grows quadratically in the state dimension. On-policy training requires every parallel environment to carry its own S_t, and off-policy training would require storing it per-transition in a replay buffer, so scaling to high-dimensional robots demands a compressed representation. Third, CIP admits degenerate maximizers, analogous to the noisy-TV problem of curiosity (Burda et al. 2018, 2019): in the pendulum systems the agent can accumulate reward by spinning freely, and in the hopper by uncontrolled thrashing. Adding joint damping removed these optima in our environments: damping taxes indiscriminate velocity accumulation while leaving configuration-dependent instability rewarded. However, this is an environment-level modification rather than an objective-level one, and a principled resolution remains open.

Each limitation suggests future work. Jacobians can in principle be extracted from a learned world model: one-step forward models of the kind trained by curiosity methods (Pathak et al. 2017) are differentiable by construction, which would make F-CIP fully learning-based and simulator-agnostic. The S_t bottleneck can be addressed with feature extractors that respect the symmetric positive-semidefinite structure of the memory, such as SPDNet (Huang and Van Gool 2017), mirroring the use of pretrained encoders for image observations in RL (Parisi et al. 2022). Finally, we plan to scale to higher-dimensional systems and, ultimately, physical robots.

A Appendix

A.1 Algorithm

Algorithm 1 outlines the full training procedure. The initial-state distribution ρ_0 is the reset distribution: the hanging configuration for the pendulum systems and the default reset for the locomotion environments. The pure-intrinsic experiment (Q2) runs only the first T_{pre} iterations, during which $\beta = 0$; the locomotion experiment (Q3) additionally runs T_{adapt} adaptation iterations with β annealed from 0 to $\beta_{\max}$ over T_{warm} iterations.

A.2 Hyperparameters

All policy and value networks are MLPs with the architecture given in Table 3; the policy outputs a Gaussian with a state-

Table 3: Hyperparameters. Environments: CP (cart pole), DP (double pendulum), TP (triple pendulum), G (gibbon), H (hopper), W (walker). T_{pre}, T_{adapt}, and T_{warm} are measured in PPO iterations, and $\beta_{\max}$ is the maximum extrinsic reward coefficient. Batch size denotes the minibatch size used for each gradient step.

	CP	DP	TP	G	H	W
Environment-specific						
T_{pre}	3500	3800	10000	10000	6000	1500
T_{adapt}	0	0	0	0	7000	7000
T_{warm}	0	0	0	0	5000	5000
$\beta_{\max}$	0	0	0	0	0.1	0.1
Rollout steps	64	64	64	256	256	256
Batch size	16384	16384	16384	32768	32768	32768
Shared across all environments						
Parallel envs	4096					
Discount γ	0.999					
GAE λ	0.95					
Clip ϵ	0.2					
Entropy coef.	0.1					
Epochs per update	2					
Learning rate	5×10^{-4}					
Optimizer	Adam					
Hidden layers	2					
Hidden dim	256					
Activation	Leaky ReLU					

dependent learned standard deviation.

A.3 Computing Resources

All experiments and model training described in this work were conducted on a single compute node equipped with two NVIDIA H100 Tensor Core GPUs and 1.0 TiB of system memory.

A.4 Environments

For the experiments involving pure CIP maximization we evaluate on four systems of increasing dimensionality: cart pole, double pendulum, triple pendulum, and gibbon, a planar humanoid-like system. These are the benchmarks of (Shah and Tiomkin 2026), with modifications: we add joint damping to the triple pendulum (3.5) and gibbon (5.0). Without damping, the policy discovers trivial maximizers of CIP, spinning the free joints to accumulate unbounded velocities rather than seeking unstable configurations. Damping bounds the attainable velocities and thereby removes these degenerate optima. We also reduce the gear strength on the original gibbon environment to 50.0.

For experiments regarding locomotion: hopper and walker, we use the default environments from the DeepMind control suite (Tunyasuvunakool et al. 2020). For both we set the internal joint damping to 1.0. Without damping we observed uncontrolled thrashing behavior as a maximizer of the CIP objective.

A.5 Training and Evaluation

All RL methods are trained under identical environmental conditions: the same environments, gear and damping settings, and the same reset distribution ($x_0 \sim \rho_0(\cdot)$). They differ only in the intrinsic objective and the underlying RL algorithm (PPO for ours, DDPG for the URLB baselines). Episodes reset to the hanging configuration in cart pole, double pendulum, and triple pendulum, so that the intrinsic reward alone must drive the exploration from the passive state to the upright one, without the agent ever being seeded there. In the gibbon environment, resets are split evenly between the hanging configuration and the vicinity of the upright stand. For hopper and walker the environments are dropped from a height above the ground with a small perturbation noise applied to its initial position.

A.6 Baselines

We compare against four objectives implemented in the Unsupervised Reinforcement Learning Benchmark (Laskin et al. 2021) with their default algorithms and parameters: DIAYN (Eysenbach et al. 2019) and SMM (Lee et al. 2020) (skill discovery/state entropy) and ICM (Pathak et al. 2017) and APT (Liu and Abbeel 2021) (curiosity / entropy-based exploration). Skill-based methods are evaluated on the skill achieving the highest average height in Table 1.

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